

# **Data Collection Methodology**

## **Research Design**

This study employs a qualitative data collection approach to examine perceptions of AI-assisted decision-making, trust, transparency, and human oversight. Participant responses are used to inform the development of a Glass Substrate Protocol (GSP), particularly in areas related to automated inspection, decision thresholds, and ethical safeguards.

## **Participants**

Participants consist of peers enrolled in coursework addressing technology, governance, and algorithmic systems. No technical background in materials science is required, as the focus is on human expectations and trust in automated systems, which are transferable to protocol design.

## **Instrument**

Data is collected through a structured set of 10 open-ended questions. These questions are designed to elicit personal experiences and opinions regarding AI and automation, while remaining aligned with the technical and ethical considerations of protocol development.

Each question corresponds to a specific protocol design consideration, such as:

- Trust thresholds
- Acceptable error rates
- Human-in-the-loop requirements
- Transparency and explainability
- Accountability and responsibility

## **Data Collection Procedure**

Participants respond to the questions in written form, either during discussion-based coursework or independently. Responses may be generated with or without the assistance of AI tools, reflecting realistic human–AI interaction environments.

Responses are anonymized using respondent identifiers and recorded verbatim to preserve meaning and nuance.

## **Data Management**

All responses are stored in a structured spreadsheet to ensure consistency, traceability, and ease of analysis. An additional column is reserved for researcher notes and thematic coding.

## **Data Analysis (Planned)**

Collected responses will be analyzed using thematic qualitative coding. Emergent themes (e.g., trust, skepticism, accountability, tolerance for error) will be mapped directly to protocol design decisions within the Glass Substrate Protocol framework.

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